A MORPHISM FROM A REDUCED SCHEME INTO A SET OF MUTUALLY INCOMPARABLE POINTS OF A LOCALLY NOETHERIAN SCHEME FACTORS THROUGH THAT DISCRETE SET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 15616-15622 and proof span 15624-15643 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof's argument that the image is contained in the finite set makes explicit use of the hypothesis that there are no nontrivial specializations among the elements of E; the author's wording is preserved.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The statement for a single point: a morphism from a reduced scheme into one point of a locally Noetherian scheme, with image exactly that point, factors through the inclusion of the point, cited from the same chapter. Cited at proof lines 15624-15626. Gives the singleton case on which the finite case is built.
- The set of minimal primes of a quotient of a Noetherian ring is finite, and the minimal primes of the image of an injective ring map are contained in the image of the corresponding morphism of spectra, cited from the algebra chapter. Cited at proof lines 15636-15640. Produces a finite subset of the given set that contains all points of the image, reducing to the finite case.

Source reference. Original lemma 00FR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 581–581. Same-chapter reference used in the retained author text at source line(s) 13632, 14321, 15637. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00FK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 574–574. Same-chapter reference used in the retained author text at source line(s) 14092, 15633. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0H1M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21252–21252. Same-chapter reference used in the retained author text at source line(s) 15626. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : Y \to X$ be a morphism of schemes. Let $E \subset X$. Assume X is locally Noetherian, there are no nontrivial specializations among the elements of E, Y is reduced, and $f(Y) \subset E$. Then f factors through $\coprod_{x \in E} x \to X$.*

Proof. When E is a singleton this follows from Lemma 0H1M. If E is finite, then E (with the induced topology of X) is a finite discrete space by our assumption on specializations. Hence this case reduces to the singleton case. In general, there is a reduction to the case where X and Y are affine schemes. Say $f : Y \to X$ corresponds to the ring map $\varphi : A \to B$. Denote $A' \subset B$ the image of φ. Let $E' \subset \mathrm{Spec}(A') \subset \mathrm{Spec}(A)$ be the set of minimal primes of A'. By Algebra, Lemma 00FK the set E' is contained in the image of $\mathrm{Spec}(B) \to \mathrm{Spec}(A') \subset \mathrm{Spec}(A)$. We conclude that $E' \subset E$. Since A' is Noetherian we have E' is finite by Algebra, Lemma 00FR. Since any other point in the image of $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is a specialization of an element of E' and in E, we conclude that the image is contained in E' (by our assumption on specializations between points of E). Thus we reduce to the case where E is finite which we dealt with above. □